$$\frac{(\mathrm{e}, l) \Downarrow_{[\![\tau_1]\!]} m \qquad (\mathrm{d}, l) \Downarrow_{[\![\tau_2]\!]} n}{(\mathrm{e}\ \mathrm{op}\ \mathrm{d}, l) \Downarrow_{[\![\mathrm{op}]\!]([\![\tau_1]\!],[\![\tau_2]\!])} [\![\mathrm{op}]\!](m,n)}\ (\mathrm{Op}) \qquad \frac{(\mathrm{i}, l) \Downarrow_{\mathbb{Z}} n}{(\mathrm{U}^f(\mathrm{i}), l) \Downarrow_{\mathbb{C}^{2\times 2}} [\![\mathrm{U}^f]\!](n)}\ (\mathrm{Unit})$$

$$\frac{}{(n, l) \Downarrow_{\mathbb{Z}} n}\ (\mathrm{Cst}) \qquad \frac{(\mathrm{s}, l) \Downarrow_{\mathcal{L}(\mathbb{N})} [x_1, \ldots, x_m] \qquad (\mathrm{i}, l) \Downarrow_{\mathbb{Z}} k \in [1, m]}{(\mathrm{s} \ominus [\mathrm{i}], l) \Downarrow_{\mathcal{L}(\mathbb{N})} [x_1, \ldots, x_{k-1}, x_{k+1}, \ldots, x_m]}\ (\mathrm{Rm}_{\in})$$

$$\frac{(\mathrm{s}, l) \Downarrow_{\mathcal{L}(\mathbb{N})} [x_1, \ldots, x_n]}{(|\mathrm{s}|, l) \Downarrow_{\mathbb{Z}} n}\ (\mathrm{Size}) \qquad \frac{(\mathrm{s}, l) \Downarrow_{\mathcal{L}(\mathbb{N})} [x_1, \ldots, x_m] \qquad (\mathrm{i}, l) \Downarrow_{\mathbb{Z}} k \notin [1, m]}{(\mathrm{s} \ominus [\mathrm{i}], l) \Downarrow_{\mathcal{L}(\mathbb{N})} [\,]}\ (\mathrm{Rm}_{\notin})$$

$$\frac{}{(\overline{\mathrm{q}}, l) \Downarrow_{\mathcal{L}(\mathbb{N})} l}\ (\mathrm{Var}) \qquad \frac{(\mathrm{s}, l) \Downarrow_{\mathcal{L}(\mathbb{N})} [x_1, \ldots, x_m] \qquad (\mathrm{i}, l) \Downarrow_{\mathbb{Z}} k \in [1, m]}{(\mathrm{s}[\mathrm{i}], l) \Downarrow_{\mathbb{N}} x_k}\ (\mathrm{Qu}_{\in})$$

$$\frac{(\mathrm{s}, l) \Downarrow_{\mathcal{L}(\mathbb{N})} [x_1, \ldots, x_m] \qquad (\mathrm{i}, l) \Downarrow_{\mathbb{Z}} k \notin [1, m]}{(\mathrm{s}[\mathrm{i}], l) \Downarrow_{\mathbb{N}} 0}\ (\mathrm{Qu}_{\notin})$$